RADINTEL AI / RADIOLOGY ONLY

Functional Documentation

Actors, business rules, functional requirements, acceptance criteria and MVP scope

Version 1.0 | 14 October 2026 | Planning draft - not clinical or legal advice

1. Product definition

RadIntel AI is a standalone radiology workflow and reporting application for diagnostic centers and radiology departments. It supports operational efficiency and physician-reviewed documentation; it is not intended in MVP to interpret medical images or produce diagnostic conclusions autonomously.

2. Actors and permissions

Actor	Primary capabilities	Key restrictions
Clinic administrator	Manage clinic configuration, users, services, templates	Cannot alter signed clinical findings
Reception/scheduler	Register patient, schedule study, manage cancellations	No access to unrelated clinical records
Radiology technician	Manage acquisition status, study metadata	Cannot sign radiologist reports
Radiologist	Review study, dictate/edit/sign report, addendum	Only assigned/authorized patients
Referring clinician	Read authorized finalized results	Cannot edit radiology reports
Compliance auditor	Inspect access logs and incidents	Read-only; least-privilege access
Platform support	Diagnose technical failures with redacted telemetry	No default access to patient content

3. Scope by release

Release	Included	Excluded
MVP (R1)	Organizations, RBAC, patients, orders, worklist, reporting, signatures, audit, DICOM study links, basic metrics	Autonomous image diagnosis, cross-institution exchange
R2	OHIF viewer integration, dictation, bilingual templates, prior-report search, secure delivery	Automated medical triage
R3	RAG over approved documents, textual longitudinal comparison, consultation, advanced analytics	Image classification without validation
R4 research	Potential image AI only with clinical study and regulatory strategy	Unapproved clinical deployment

4. Core business objects

Organization; Facility; User; Role; Patient; PatientIdentifier; Referrer; ImagingOrder; Appointment; Modality; ImagingStudy; SeriesReference; WorklistAssignment; Report; ReportVersion; ReportTemplate; DictationTranscript; AIProposal; Signature; Addendum; Notification; AuditEvent; Equipment; DowntimeEvent; ConsentOrLegalBasisRecord.

5. Functional requirements (FR)

FR-001 - Organization onboarding

Requirement: Create separate clinic tenant with facilities, modalities, staff and locale.

Acceptance criteria: Admin creates tenant; users from tenant B cannot see A.

FR-002 - Identity and access

Requirement: Authenticate staff with MFA and enforce role and patient-scope permissions.

Acceptance criteria: Unauthorized resource requests return denied and are logged.

FR-003 - Patient registration

Requirement: Register patient, validate identifiers, detect potential duplicates.

Acceptance criteria: Duplicate warning; merge only by privileged verified workflow.

FR-004 - Imaging orders

Requirement: Create exam order with requested modality, body region, referral metadata.

Acceptance criteria: Required fields validated; status changes tracked.

FR-005 - Scheduling

Requirement: Book, move, cancel appointments; track reason and priority assigned by clinician.

Acceptance criteria: No overlap beyond configured scanner capacity; audit changes.

FR-006 - Worklist

Requirement: Show scheduled, acquired, in-review, pending-signature and finalized studies.

Acceptance criteria: Transitions are role-constrained and timestamped.

FR-007 - PACS linkage

Requirement: Link imaging study by StudyInstanceUID and archive endpoint.

Acceptance criteria: Viewer link resolves only for authorized users.

FR-008 - Radiology reporting

Requirement: Create draft using modality/body-region template; save revisions.

Acceptance criteria: Unsigned drafts clearly marked and never delivered as final.

FR-009 - Dictation assistant

Requirement: Transcribe physician dictation and propose formatted draft.

Acceptance criteria: Doctor sees raw transcript and approves every proposed edit.

FR-010 - AI provenance

Requirement: Store AI model, prompt/template version, source references and reviewer action.

Acceptance criteria: Each AI draft traceable to version and approving clinician.

FR-011 - Report signing

Requirement: Sign final report, freeze version and record signature details.

Acceptance criteria: Signed content immutable; correction requires addendum.

FR-012 - Longitudinal history

Requirement: Display previous approved reports and authorized imaging studies.

Acceptance criteria: Patient matching verified; cross-patient leakage tests pass.

FR-013 - Secure distribution

Requirement: Send finalized report through authenticated portal or expiring access link.

Acceptance criteria: Links expire/revoke; downloads audited.

FR-014 - Search

Requirement: Filter orders/reports by permitted patient, date, modality, facility and status.

Acceptance criteria: Pagination, access control and predictable performance.

FR-015 - Analytics

Requirement: Display counts, backlog, median turnaround, no-show and utilization.

Acceptance criteria: Metrics reconciled with underlying audit events.

FR-016 - Audit and retention

Requirement: Log sensitive actions and enforce clinic-specific retention policies.

Acceptance criteria: Immutable audit trail and tested export/retention workflows.

FR-017 - Localization

Requirement: Support Albanian and English UI and report templates.

Acceptance criteria: Clinical terminology validated by local radiologists.

FR-018 - System integration

Requirement: Support configured DICOMweb and future FHIR adapters.

Acceptance criteria: Contract tests and clear failure/retry states.

6. Workflow state machines

Imaging order: Draft -> Requested -> Scheduled -> CheckedIn -> Acquired -> ReadyForReview -> InReview -> ReportDraft -> Signed -> Delivered. Alternate paths: Cancelled, Rescheduled, AcquisitionFailed, CorrectionPending. Each transition has allowed actor, timestamp, reason, and idempotency key.

Report: Draft -> PendingReview -> Signed. Post-signature corrections produce a linked Addendum record; they do not overwrite the signed original.

7. Non-functional requirements (NFR)

Area	Initial target / control
Availability	Pilot target 99.5% monthly excluding agreed maintenance; refine per contract
Performance	P95 ordinary metadata API < 500 ms under defined pilot load; image delivery depends on PACS/network
Security	TLS, encryption at rest, MFA, tenant isolation, secrets management, audit logs
Privacy	Data minimization, lawful basis, access logging, retention/deletion policy, DPA
Recovery	Illustrative RPO <= 24h and RTO <= 8h for pilot; test restore drills
Accessibility	Keyboard navigation, readable contrast, error messages
Localization	Albanian and English; terminology reviewed by clinicians
Observability	Correlation IDs, trace sampling, alerts without patient content

8. AI guardrails and human oversight

- Every generated report remains explicitly DRAFT until a qualified radiologist verifies and signs.
- Use retrieval grounding for policy/template suggestions; do not manufacture observations absent from dictated notes.
- Never auto-populate clinical findings from images without separately validated image AI.

- Log prompts and responses only where privacy-approved; default to redacted metadata and protected access.
- Provide feedback flag, rollback, AI-off switch and error escalation.

9. Pilot acceptance checklist

- End-to-end workflow demonstrated with synthetic studies and patient records.
- Threat model reviewed, tenant isolation tested, and clinical safety sign-off documented.
- Clinic signs off on templates, report format, access controls, training and incident response.
- Performance baselines and success metrics agreed before pilot start.

References and evidence boundaries

World Bank, Kosovo healthcare strengthening (2024) — https://www.worldbank.org/en/news/press-release/2024/05/07/world-bank-supports-enhancement-of-health-services-in-kosovo-improving-the-lives-of-its-people

Kosovo Ministry of Health, IHIS oversight procurement (2026) — https://msh.rks-gov.net/Documents/Index/1?year=2026

World Bank, Kosovo Digital Health Blueprint case study — https://thedocs.worldbank.org/en/doc/b929829fc3d36bebdf9d93555c64f222-0140022024/related/Digital-Health-Blueprint-Toolkit-Kosovo-Case-Study.pdf

HL7 FHIR specification — https://hl7.org/fhir/

OHIF documentation — https://docs.ohif.org/

OHIF DICOMweb and Orthanc integration — https://docs.ohif.org/configuration/datasources/dicom-web/

Market demand, exact waiting times, clinic counts, current vendor penetration, willingness to pay, and legal classification require local primary research. All numerical commercial scenarios in this document are illustrative assumptions.